The **Rollback Methods** parameters determine the rollback procedures used to simulate out-of-attainment monitors coming into attainment:

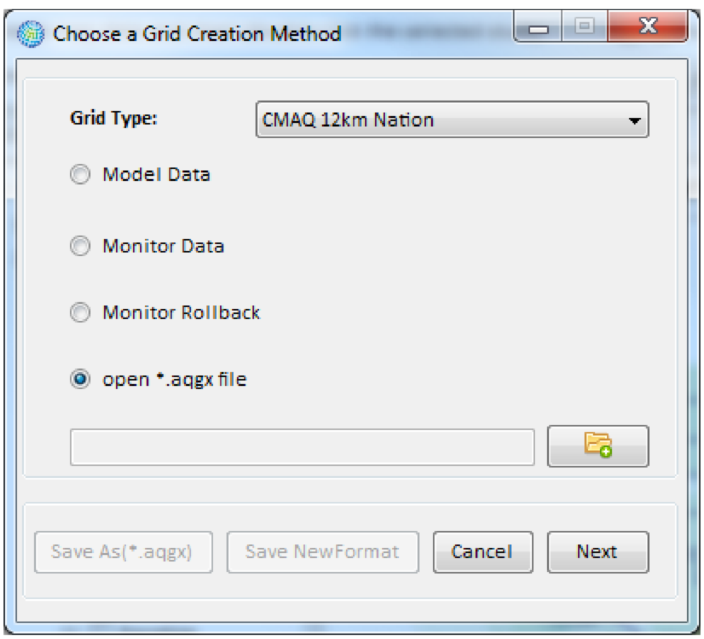
- **Interday Rollback Method** (with associated **Background** level) – These are used to generate target values for the metric specified by the **Attainment Test**. Method types include *Percentage*, *Incremental*, and *Peak Shaving.*

- **Intraday Rollback Method** (with associated **Background** level) **–** These are used to adjust hourly observations to meet the target metric values generated in the previous step. Method types include *Percentage* and *Incremental.*

The methods involved for each can be somewhat complicated, so we have included a section in Appendix A: Monitor Rollback Algorithms which goes through several examples.

5.4 Open *.aqgx File

The final option for uploading an **Air Quality Grid** is to select the *Open *.aqgx File* option from the **Choose a Grid Creation Method** window.

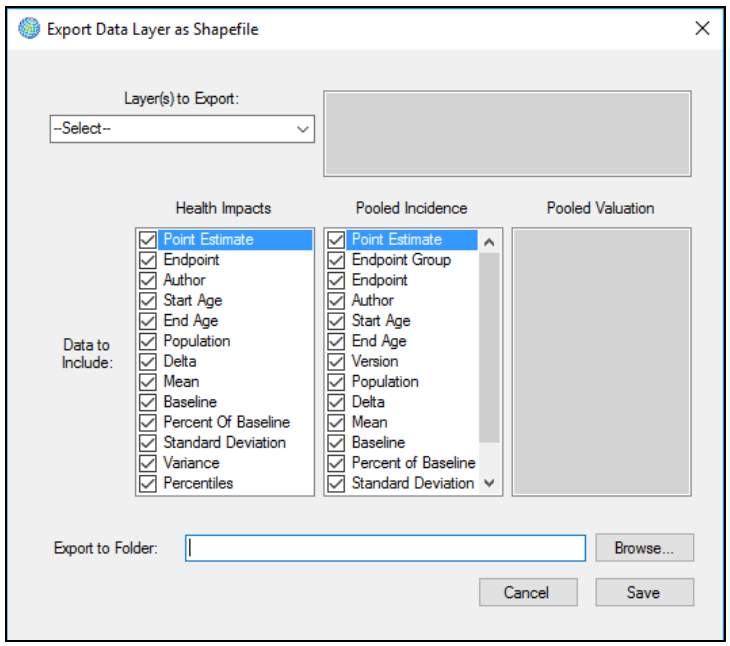


Choosing this option will activate the **Open File Browser** button located directly below this option. Click the **Open File Browser** button. This will cause an **Open** window to appear, allowing you to search for an **Air Quality Grid** (*.aqgx file) that has already been created. Select a file and click **Open**. The file path and name should appear in the box beside the **Open File Browser** button. Click **Next** to close the window and begin to create the map layer.

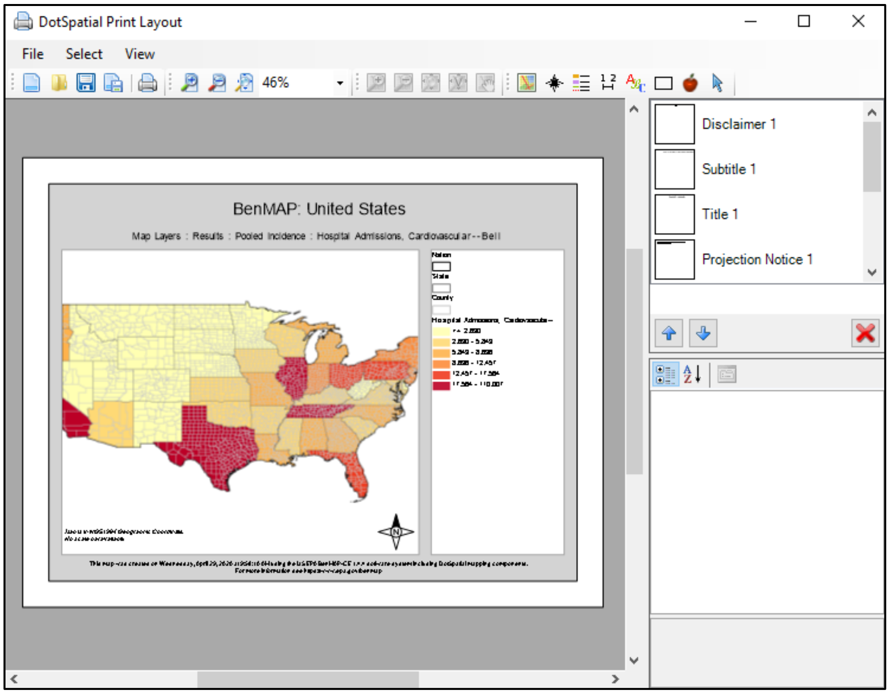
5.5 Frequently Asked Questions

How can I generate a map of an air quality grid and export it?

When viewing any of the displayed maps n the **GIS Map** tab (lower right frame of the main window), click on the GIS toolbar icon for **Save Shapefile** (looks like a 3.5-inch diskette). You can choose the Layer(s) to Export from the dropdown menu and select data to include in the exported shapefile. Follow the prompts to provide a name and location for the file. BenMAP-CE will export a set of files (.dbf, .prj, .shp, .shx) associated with the shapefile that you can use with any GIS viewer.

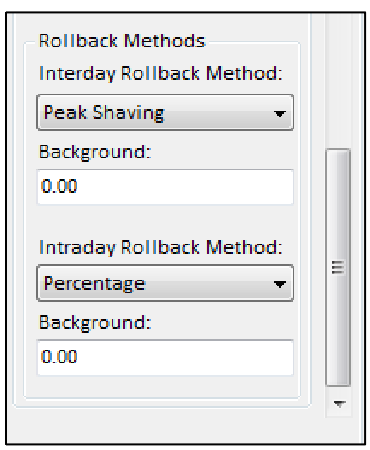


To export the map as an image, click the **Prepare Print Layout** icon (immediately above the **Save Shapefile** icon). This will use built-in DotSpatial GIS tools to allow you to save the map as a formatted image (.png) file.



For the Rollback to a Standard option, why are there Interday and Intraday rollback options?

The **Interday Rollback Method** option identifies the approach (e.g., *Percentage*) to reduce daily air pollution levels, in order to meet the specified standard. (In other words, there is more than one way to reduce daily pollution levels so as to meet the standard you have chosen, and BenMAP-CE lets you choose from among those approaches.) The **Background** level associated with the **Interday Rollback Method** specifies the bound, below which, BenMAP-CE will not make adjustments to daily levels.



The **Intraday Rollback Method** option is only relevant for hourly air pollution data, like ozone measurements. This option specifies the approach (e.g., *Percentage*) used by BenMAP-CE to reduce hourly air pollution levels to reach the target metric values. That is, once you have chosen the approach to reduce daily air pollution levels, on any given day there is more than one way to reduce the hourly air pollution values to meet the targeted pollution level for that day. The **Background** level associated with the **Intraday Rollback Method** specifies the bound, below which, BenMAP-CE will not make adjustments to hourly levels.

The **Interday** and **Intraday** options are complicated. Appendix A on Monitor Rollback Algorithms explains these options in more detail and gives some numerical examples.

Can I use air quality grids based on different Grid Types in the baseline and control scenarios?

No. In any given analysis, you need to use the same Grid Type in the baseline and control scenarios.

Can I use air quality grids of the same Grid Type but based on different Grid Creation Methods?

Yes. In any given analysis, you may use air quality grids made with different methods. Air quality grids made with Model Data and Monitor Data may be used interchangeably, if desired. Similarly, air quality grids made with different interpolation methods may be compared. However, it generally is not recommended to create grids with different methods and use them in the same analysis.

Can I do an analysis with multiple pollutants?

You can currently only estimate impacts one pollutant at a time; however, BenMAP-CE allows you to aggregate the results of more than one pollutant. This is discussed in Chapter 7: Aggregating, Pooling, and Valuing.

Why does it take so long to generate an Ozone Air Quality grid if there are a lot of grid cells?

It can take a long time to create an air quality grid because the file being generated can be quite large. In some cases, air quality grids can be several hundred megabytes in size. (One reason the ozone files are large is that the definition of ozone has, by default, four metrics. If you do not need all of the default metrics for the health impact functions in your database, then delete the unneeded metrics and BenMAP-CE will run faster and generate smaller air quality grids. This is an advanced step, so do not do it if you are unsure.) The type of computer you use can also affect processing speeds. Refer to Chapter 1, Section 1.3 for recommended hardware specifications.

How do I access data in an Air Quality Surface?

You can access the data in an air quality grid by going to the **Tools** menu and choosing the **Export Air Quality Surface** option. (The **Tools** menu is available on the toolbar of the main BenMAP-CE window.) Locate the air quality grid from which you want to